

BREAK INTO AEROSPACE

GNSS Engineering

Subdomain Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

Global Navigation Satellite Systems is one of the most technically demanding niches in aerospace engineering — a field that sits at the intersection of orbital mechanics, radio frequency engineering, statistical signal processing, control theory, and applied geodesy. It is also one of the most commercially consequential: GNSS underpins aviation safety, autonomous vehicles, precision agriculture, financial market timestamps, mobile mapping, and military targeting simultaneously. Engineers who understand GNSS at depth are in persistent demand across defence primes, chipset vendors, receiver manufacturers, aviation system integrators, and space agencies.

This guide is structured for engineers at all levels targeting roles in GNSS receiver design, navigation algorithm development, augmentation systems, inertial integration, signal processing, or GNSS infrastructure — at employers as different as u-blox (Swiss chipset vendor), NovAtel (Canadian high-precision receivers), Septentrio (Belgian geodetic receivers), Spirent Communications (UK test and simulation), Raytheon (US GNSS infrastructure and defence), Collins Aerospace (aviation navigation systems), Thales Alenia Space (Galileo ground segment), and ESA's Navigation Directorate.

PART I · WHAT THE GNSS FIELD ACTUALLY LOOKS FOR

By Level

- **Intern / Graduate** — At entry level, GNSS employers are not primarily screening for GNSS-specific knowledge — they are screening for signal processing fundamentals, mathematical rigour, and the ability to connect theoretical concepts to real system behaviour. The distinction between a strong and a weak graduate candidate is not usually domain knowledge (which can be taught) but mathematical intuition: does this person understand what a Kalman filter is doing, not just how to invoke one? Can they derive a least-squares position solution from scratch, or only run a software library? Can they explain why carrier phase is more precise than pseudorange, and what ambiguity resolution requires?

Graduate interviews at larger employers — Collins Aerospace, Honeywell, L3Harris, Thales, Raytheon — typically include a technical screen covering signal processing, linear algebra, and probability, followed by one or two interviews combining behavioural questions with domain probing at conceptual depth. At smaller specialist employers — Septentrio, Swift Navigation, Rokubun, u-blox — the screen is often a take-home problem or coding exercise involving GNSS data (RINEX files, raw measurements, or simulated signals), followed by a technical deep-dive with a senior engineer. What distinguishes graduate candidates is evidence of genuine engagement with GNSS beyond coursework: a thesis using real GNSS data, a software receiver project, or a well-documented personal project using low-cost receivers.

- **Mid-Level (Engineer, Senior Engineer, ~3–8 years)** — At mid-level, GNSS interviews become product and problem specific. Interviewers are not assessing conceptual understanding — they are assessing whether you can solve real problems in this domain with real constraints. The questions shift from “explain how a Kalman filter works” to “how would you handle code-minus-carrier divergence in a dual-frequency SBAS monitor?” or “what happens to your position solution when a single satellite transitions from above to below the mask angle during a tight coupling integration?”

Technical interviews at mid-level commonly include whiteboard or screen-share problems where the candidate must reason through an error budget, a filter design trade-off, or a receiver architecture decision. Coding exercises involving navigation data processing — parsing RINEX, implementing a weighted least-squares PVT solver, or writing a simple tracking loop discriminator — are standard at software-defined receiver or algorithm development roles. The specific context driving mid-level hiring in 2025–2026: the LEO navigation augmentation sector is in early but genuine acceleration (Xona Space Systems, Satelles, Galileo HAS expansion), automotive autonomy requires centimetre-level GNSS robust to urban environments, European defence investment is driving demand for assured PNT, and the opening of Galileo High Accuracy Service and BeiDou PPP-B2b at no cost is reshaping the precision agriculture and surveying market.

- **Senior (Principal Engineer, Technical Lead, Fellow)** — At senior level, GNSS employers are looking for engineers who own complete technical threads — from system requirements through algorithm design through verification — and who can make defensible architectural choices when the constraints are competing and the answer is not obvious. The interview is often structured as a technical discussion rather than a test: what is your position on the trade-off between loose and tight coupling for automotive applications where GNSS outages are frequent but IMU drift accumulates? How would you architect a fault detection system for a dual-constellation SBAS when the two constellations have correlated atmospheric threats? What does the move to LEO augmentation mean for traditional RAIM algorithm design?

Senior roles at navigation system integrators (Collins, Honeywell, Thales), space agencies (ESA, NASA JPL), research institutions (DLR, ETH Zurich PNTL, Stanford GPS Lab), and precision agriculture platform companies require engineers who can engage at this level without preparation time. The distinguishing credential at senior level is having shipped a GNSS-dependent system to production — not just having designed algorithms, but having taken them through verification, certification (DO-178C for aviation, ISO 26262 for automotive, or equivalent), and operational use.

Hands-On Experience and Domain Knowledge Treated as Strong Signals

Domain	Why It Signals Depth
Software-Defined Receiver Implementation	Engineers who have built a software GNSS receiver — even partial, even in MATLAB or Python — from raw IF samples through acquisition, tracking, and navigation data decoding demonstrate depth that no amount of textbook study can substitute. The canonical reference is Borre et al., <i>A Software-Defined GPS and Galileo Receiver</i> (Birkhäuser, 2007).
Kalman Filtering — Design, Not Application	“I have used Kalman filters” is a weak signal. “I designed the process noise matrix for a pedestrian dead reckoning application, observed filter divergence under a specific dynamics scenario, and diagnosed it as a Q-matrix misconfiguration” is a strong signal.
GNSS Error Modelling — All Sources, Quantitative	Engineers who can state the typical magnitude of each error source in metres, explain which are correlated across nearby receivers versus uncorrelated, and describe the modelling approaches (Klobuchar, MOPS troposphere, NEQUICK) are demonstrating domain literacy specifically tested at mid-level and above.
Augmentation System Architecture	WAAS, EGNOS, MSAS are mature; Galileo HAS, BeiDou PPP-B2b, QZSS CLAS are operational; PPP-RTK systems from Trimble, Leica, and Hexagon are commercially active. Understanding what each provides in accuracy, convergence time, and integrity characterisation is valuable across the industry.
Integrity Monitoring — RAIM, ARAIM, FDE	Aviation GNSS depends on receiver integrity monitoring. Engineers with experience designing or evaluating RAIM algorithms — understanding protection level computation, snapshot versus sequential methods, and the integrity-continuity-availability trade-off — are specifically relevant to aviation integrators and SBAS developers.
INS/GNSS Integration — Coupling Architectures	Loose (position/velocity aiding), tight (pseudorange/Doppler aiding), or deep (correlator-level integration) coupling architectures determine outage performance, computational complexity, and fault isolation properties.
C++ or Python with Numerical Methods Fluency	Engineers fluent in both production firmware (C/C++) and analysis/prototyping (Python with NumPy/SciPy, MATLAB) who can implement a complete PVT solver from scratch are employable at algorithm engineering level at almost any GNSS company.
Survey or Geodetic GNSS Experience	Centimetre-level positioning requires carrier phase observables and integer ambiguity resolution (LAMBDA method, FARA, MLAMBDA). Engineers with RTK or PPP implementation experience are specifically relevant to surveying, precision agriculture, and maritime navigation markets.
Simulation and Test Tooling	Experience with GNSS signal simulators (Spirent GSS7000/9000, R&S SMBV100B, Keysight E4438C) to characterise receiver performance under interference, signal blockage, or spoofing is a practical skill specifically valued at receiver manufacturers

How GNSS Interviews Differ from Generic Aerospace or Software Roles

- **The mathematics is unavoidable.** GNSS is one of the most mathematically intensive applied engineering disciplines. Linear algebra (the PVT least-squares problem is a matrix inversion), probability theory (the Kalman filter is a Bayesian estimator), signal processing (correlation, spectral analysis, discriminator design), and differential geometry (coordinate frames, attitude representation) all appear in a typical GNSS engineer's working week. Interviews that do not include mathematical reasoning are not GNSS interviews.
- **Domain knowledge is tested at specificity, not breadth.** Interviewers do not ask "what is GNSS?" They ask things like: "if the signal-to-noise ratio drops by 6 dB, what happens to your pseudorange measurement noise, and how does this propagate into your position solution?" The expected answer is not an outline — it is a technically specific explanation at engineering depth.
- **Coding exercises are problem-specific, not algorithm-generic.** GNSS software companies use take-home or live coding exercises involving real GNSS data — not generic LeetCode-style exercises. A typical exercise: parse a RINEX 3 observation file, compute satellite positions from broadcast ephemeris using the IS-GPS-200 algorithm, compute a pseudorange solution with iterative least squares, and analyse the residuals for evidence of a faulty measurement. The candidate who has done this before completes it in hours; the candidate who has not may struggle with the file format alone.
- **Clearance requirements are real for defence roles.** GPS P(Y) and M-code receivers, anti-jam arrays, and secure timing devices require security clearance at the appropriate national level (TS/SCI in the US, DV in the UK, NATO-equivalent in other nations). Raytheon, L3Harris, BAE Systems Navigation, and their national equivalents all have GNSS roles that are clearance-gated. US citizenship is typically required for US roles; British nationality or long-term UK residency is assessed by UKSV for UK roles.
- **The open-source community provides a meaningful signal.** GNSS-SDR (an open-source software-defined GNSS receiver in C++), RTKLIB (the canonical open-source RTK/PPP processing library), and contributing to these projects or demonstrating fluent use of them for a specific analysis is a credible signal of technical depth that employers recognise. Academic conference papers at ION GNSS+, ENC (European Navigation Conference), or the ION Pacific PNT conference carry weight even for industry candidates.
- **The field is genuinely interdisciplinary — and interviews probe the seams.** A signal processing specialist who cannot explain why geometric dilution of precision matters is missing a connection their algorithms directly affect. An algorithm engineer who cannot explain what the receiver noise floor means for their filter's measurement noise matrix has an abstraction gap their colleagues will notice.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern / Graduate

1. Build a Complete Pseudorange PVT Solver from Scratch — in Python, Against Real GNSS Data 8–12 h

Skill: Implementing the core GNSS positioning problem end-to-end

Objective: The position-velocity-time solution from pseudorange measurements is the foundational computation of GNSS. Every algorithm built on top of it — RAIM, differential corrections, tight coupling — depends on understanding what is happening here. A candidate who has implemented this from scratch — parsing RINEX 3 observation and navigation files, computing satellite ECEF positions from Keplerian elements using the IS-GPS-200 algorithm, applying ionospheric and tropospheric corrections, forming the H matrix and weight matrix, and solving iteratively until convergence — understands GNSS from the ground up. The distinction between having done this and

having only used a positioning library is real and is detectable in interviews within two or three questions. 8–12 h

Done Signal: You have a working Python script — ideally on GitHub — that reads a RINEX 3 observation file and a broadcast navigation file, computes satellite positions, applies at least a Klobuchar ionospheric correction and a simple tropospheric correction, solves for receiver position and clock bias by iterated weighted least squares, and plots the resulting horizontal position error against a known reference. You can explain every line of the computation without referring to the code.

2. Understand the Four Major GNSS Constellations — Signal Structures, Frequencies, and What Distinguishes Them 4–5 h

Skill: Constellation literacy as the baseline of GNSS domain knowledge

Objective: Modern GNSS receivers are multi-constellation (GPS, GLONASS, Galileo, BeiDou) and multi-frequency (L1/E1, L2/E6, L5/E5). Interviewers at any GNSS employer will probe constellation-level knowledge directly: why does Galileo’s E1 Open Service signal have a different chip rate from GPS L1 C/A? What are the orbit characteristics of BeiDou’s GEO, IGSO, and MEO satellites? What makes GLONASS’s FDMA signal structure different from GPS’s CDMA, and why does this complicate inter-constellation integer ambiguity resolution? These are baseline domain knowledge for a practising GNSS engineer.

Done Signal: For GPS, GLONASS, Galileo, and BeiDou: you can state the primary civil signal frequencies and bandwidths, the orbit type and altitude, the number of operational satellites, and one specific technical property that distinguishes this constellation from the others. You can explain what the transition to CDMA means for GLONASS’s next-generation satellites and why this matters for receiver design.

3. Understand the Kalman Filter as a Bayesian Estimator — Derive the Update Equations, Not Just Apply Them 4–6 h

Skill: Statistical estimation at the mathematical depth GNSS interviews demand

Objective: GNSS navigation uses Kalman filtering as its standard estimation framework. The canonical interview question is “walk me through the Kalman filter update step” — and the expected answer involves the Kalman gain, the innovation covariance, and why the form $K = PH^T(HPH^T + R)^{-1}$ minimises the mean squared error. Candidates who can derive this from the Bayesian update equations for a Gaussian prior and Gaussian likelihood are demonstrating the mathematical depth the field requires. Candidates who recite the equations without understanding their origin are demonstrating memorisation — detectable in two follow-up questions.

Done Signal: You can derive the Kalman filter update equations from the Bayesian perspective — starting from the Gaussian prior on state and the Gaussian measurement likelihood, deriving the posterior mean and covariance, and arriving at the Kalman gain expression. You can explain what the process noise covariance Q represents physically, why choosing Q too small causes filter lag, and why choosing it too large causes filter noise amplification.

4. Prepare Behavioural Answers That Connect Technical Work to System Impact 2–3 h

Skill: STAR-format stories for a technically demanding field

Objective: GNSS employers — especially aviation system integrators, SBAS developers, and precision instrument companies — probe collaboration under technical ambiguity, behaviour when a system performs unexpectedly in test, and willingness to acknowledge and correct a wrong tech-

nical position. Stories that end with “the system worked as designed” are weaker than stories that end with “we found the error mode in hardware-in-the-loop testing before it reached the flight test programme.”

2–3 h

Done Signal: You have four STAR stories covering: a technical problem you diagnosed and solved using data, a time you worked across a disciplinary boundary (signal processing and RF hardware, or algorithm and firmware), a quality or verification step you personally introduced that caught a real failure mode, and a failure or wrong technical decision you owned and recovered from. Each story ends with a specific, quantified outcome where possible.

5. Work Through the GNSS Error Budget Quantitatively — All Sources, All Magnitudes

3–4 h

Skill: The ability to reason about positioning accuracy without running a simulation

Objective: GNSS accuracy analysis starts with an error budget: a breakdown of all error sources contributing to the pseudorange measurement, their typical magnitudes, their correlation properties, and their sensitivity to mitigation strategies. Interviewers at every level probe this. “If I switch from single-frequency to dual-frequency processing, which error sources do I reduce, and by how much?” requires knowing both that dual-frequency processing eliminates the first-order ionospheric delay and that the typical ionospheric delay at L1 is 5–15 metres, occasionally exceeding 50 metres during solar maxima.

Done Signal: For each of the following GNSS error sources, you can state the typical magnitude in metres (pseudorange equivalent), the dominant mitigation approach, and whether the error is correlated across nearby receivers: ionospheric delay (single-frequency), tropospheric delay, satellite clock error, broadcast ephemeris error, multipath, receiver noise, and antenna phase centre variation. You can explain what UERE (User Equivalent Range Error) means and how it relates to PDOP and the resulting position accuracy.

6. Get Hands-On with Real GNSS Data — RINEX Files, CORS Networks, or Raw Measurements from a u-blox or Android Device

4–6 h

Skill: Practical data familiarity that distinguishes engaged candidates in screening

Objective: Real GNSS data is freely available. The IGS network and national CORS networks provide observation and navigation data for thousands of reference stations worldwide. u-blox modules connected to a laptop expose raw GNSS measurements over UBX protocol. Android phones with GNSS raw measurement support (available since Android 7) expose carrier phase, pseudorange, and Doppler via the `GnssRawMeasurement` API. Candidates who have processed any of these data sources — even informally, even with RTKLIB — have a practical familiarity that is immediately evident in conversation: they know what cycle slips look like in the data, what a poor PDOP epoch means for solution quality, and what an outlying pseudorange residual suggests.

Done Signal: You have downloaded or collected real GNSS observation data (from IGS, a CORS station, a u-blox module, or an Android device), processed it using either your own code or RTKLIB, and can describe at least two specific observations from the data: an anomaly you noticed and diagnosed, a satellite signal behaviour you found interesting, or a performance characteristic you quantified. You know what RINEX 3 looks like and can navigate its structure without consulting documentation.

Mid-Level (~3–8 years)

All graduate tasks still apply. Behavioural stories must come from professional contexts. Coding ability must extend to production-quality C++ or Python, not just prototyping.

7. Understand SBAS Architecture — from Monitor Station to Airborne Receiver, Including Integrity Computation 6–8 h

Skill: Safety-critical augmentation system engineering from message generation to integrity assurance

Objective: Satellite-Based Augmentation Systems — WAAS in the US, EGNOS in Europe, MSAS in Japan, GAGAN in India — are the aviation industry’s mechanism for delivering GNSS with certified integrity for approach procedures. The architecture spans a ground network of reference stations, master stations that compute differential corrections and integrity bounds, an uplink to a geostationary satellite, and an airborne receiver that applies corrections and computes a protection level. Engineers working at SBAS ground segment developers (Raytheon WAAS, Thales EGNOS, GMV, INDRA) or airborne receiver manufacturers (Collins, Honeywell, Garmin Aviation) are expected to understand the full chain — specifically how UDRE (User Differential Range Error) and GIVE (Grid Ionospheric Vertical Error) bounds are computed. This requires reading RTCA DO-229 and EUROCAE ED-144.

Done Signal: You can explain what an SBAS message type 2 (fast corrections) and type 26 (ionospheric grid point corrections) contain, how the UDRE and GIVE values are derived from reference station measurements, what the VUFP (Vertical Urban Fault Position) concept means for EGNOS threat model expansion, and how the airborne receiver computes a vertical protection level from these inputs. You have read the relevant sections of DO-229 or ED-144.

8. Prepare a Technically Specific Explanation of INS/GNSS Integration — Loose, Tight, and Deep Coupling 5–6 h

Skill: The standard architecture for continuous navigation in GNSS-degraded environments

Objective: Inertial/GNSS integrated navigation is the design solution whenever GNSS alone is insufficient — in urban canyons, tunnels, denied environments, or during signal outages. The coupling architecture determines what the filter ingests as measurements: position and velocity (loose), pseudorange and Doppler (tight), or correlator in-phase and quadrature outputs (deep/ultra-tight). Tight coupling can continue providing position updates with fewer than four satellites visible; deep coupling can maintain tracking continuity in jamming environments that defeat conventional tracking loops. Engineers who can explain these trade-offs and describe which architecture they have implemented are demonstrating applied depth.

Done Signal: You can explain what changes in the state vector, measurement model, and H matrix when transitioning from loose to tight coupling INS/GNSS integration. You can describe what the IMU error state vector typically contains (accelerometer bias, gyroscope bias, scale factor errors), why IMU errors are modelled as random walks rather than white noise, and how the filter error state is used to correct the mechanisation solution. You can name one specific application where you would choose deep coupling over tight coupling and justify the choice.

9. Build Working Knowledge of Carrier Phase Observables — What They Are, Why They Are Precise, and What Integer Ambiguity Resolution Requires 5–7 h

Skill: The technical foundation of centimetre-level GNSS positioning

Objective: Pseudorange positioning reaches 1–3 metre accuracy under good conditions. Carrier phase positioning reaches centimetre accuracy — but requires resolving the integer ambiguity in the phase observable, the unknown number of complete carrier wavelengths between satellite and receiver at initial lock. The LAMBDA method and its derivatives (MLAMBDA, partial ambiguity resolution) are standard. The interview question “explain how RTK works and what limits its baseline length” tests whether the candidate understands that carrier phase double differences remove satellite and receiver clock errors, that the ionosphere and troposphere limit how far the baseline

can extend, and that the ambiguity can be fixed for baselines under approximately 30 km without ionospheric modelling. 5–7 h

Done Signal: You can explain the formation of a double-differenced carrier phase observable and which error sources it eliminates. You can describe what the LAMBDA method does — stating that it decorrelates the ambiguity covariance matrix before searching and why this matters for search efficiency — and explain the ratio test used to validate a fixed solution. You can state the typical baseline length limit for RTK without atmospheric modelling and explain what PPP-AR requires that RTK does not.

10. Prepare a Story About Diagnosing an Unexpected Navigation System Failure Using Data — Not Simulation 2 h

Skill: Systematic fault diagnosis in a deployed or tested GNSS system

Objective: GNSS systems fail in ways that simulations rarely predict: multipath from a new building, a firmware update introducing a subtle tracking loop regression, an ionospheric storm degrading solution accuracy during a flight test window, a specific satellite geometry causing a RAIM exclusion the system was not designed to handle gracefully. Engineers who have diagnosed a real system failure using actual navigation data — receiver logs, PVT logs, raw measurement logs, or reference system comparisons — demonstrate applied judgement that is not replicable from textbooks. The diagnostic process — hypothesis formation, data mining, isolation of the failure mode, root cause identification, and design change — is the core engineering skill at mid-level.

Done Signal: You have a story where a GNSS-dependent system behaved unexpectedly, you diagnosed the cause using measurement or log data, identified the root cause (a hardware issue, an algorithm edge case, an environmental effect, a configuration error), and drove a specific corrective action. The story names the system, the anomaly, the data you examined, the hypothesis you formed, how you confirmed it, and what changed as a result.

11. Understand GNSS Interference — Jamming, Spoofing, and the Engineering Response to Each 4–5 h

Skill: Assured PNT — the central technical challenge of GNSS in contested environments

Objective: GNSS vulnerability to intentional interference is no longer an academic concern. Commercial drone jammer availability, maritime spoofing incidents in the Black Sea and Eastern Mediterranean, and demonstrated GPS spoofing of aircraft in conflict zones have moved interference mitigation from a defence specialisation to a requirement across aviation, maritime, and autonomous vehicle applications. The engineering responses differ by threat: controlled reception pattern antennas (CRPAs) and null-steering for jamming; authentication signals (Galileo OSNMA, GPS Chimera) for spoofing detection at signal level; consistency monitoring (clock, position, velocity checks) for spoofing detection at navigation level.

Done Signal: You can explain the physical mechanism of a swept-tone jammer and why it is effective against spread-spectrum GNSS signals at low power. You can describe what GPS spoofing requires from an attacker (signal structure knowledge, synchronisation, trajectory plausibility) and why civilian GPS signals are more susceptible than military M-code. You can describe three mitigation approaches — one antenna-based, one signal-authentication based, and one navigation-data consistency based — and state what threat model each addresses.

12. Study RAIM and ARAIM — Protection Level Computation and the Integrity-Continuity-Availability Trade-Off 5–6 h

Skill: Navigation integrity monitoring — the technical basis of aviation GNSS safety certification

5–6 h

Objective: Aviation GNSS approval depends on integrity monitoring: the system must detect a hazardously misleading position output before it causes an accident, with a probability guarantee defined in the applicable TSO (TSO-C145/C146 for SBAS receivers, TSO-C196 for ARAIM). RAIM algorithms compute a protection level and compare it against an alert limit; if the protection level exceeds the alert limit, the system declares GNSS unavailable for the intended operation. ARAIM extends this to dual-constellation operation, requiring new fault models and new assumptions about constellation-level fault probability. Engineers at Collins Aerospace, Honeywell Aerospace, Garmin Aviation, and SBAS ground segment developers are required to understand this framework at implementation depth.

Done Signal: You can explain the snapshot RAIM algorithm: computing the test statistic, the threshold, and the protection level, and explaining what each depends on. You can describe the key difference between RAIM and ARAIM — the addition of constellation-level fault modes to the fault hypothesis set — and state what MERR (Maximum Effective Range Rate) means in this context. You can explain the three-way trade-off between integrity risk, continuity risk, and availability, and give one example of a requirement that tightens one at the cost of the others.

6–8 h

13. Prepare for a Live Coding Exercise Involving GNSS Data Processing — RINEX Parsing, Least-Squares, Residual Analysis

Skill: Producing working code on a GNSS problem under interview conditions

Objective: GNSS algorithm companies conduct live or take-home coding exercises using real GNSS data formats. The typical exercise — parse a RINEX file, compute satellite positions, form pseudorange measurements, solve by weighted least squares, analyse residuals — is entirely tractable for an engineer who has done it before. For an engineer who has not, unfamiliarity with RINEX format structure, the Keplerian propagation algorithm (which involves an iterative solution for eccentric anomaly), and the ECEF coordinate transformation causes enough friction to meaningfully reduce performance under time pressure.

Done Signal: You can implement, in under 90 minutes, a Python function that reads a RINEX 3 observation file and a broadcast navigation file, computes GPS satellite positions in ECEF at specified epochs using the IS-GPS-200 orbit model, and forms corrected pseudorange measurements. You have done this at least once before the interview, without time pressure, and you know where the iterative steps are and how to handle the edge cases (crossing week boundaries in time, Earth rotation correction during signal transit time).

Senior Level (Principal Engineer, Technical Lead, Fellow)

All mid-level tasks still apply. Kalman filter knowledge must extend to implementation trade-offs, numerical conditioning, and filter tuning under real-world non-Gaussian noise. RTK/PPP knowledge must include practical experience with ambiguity resolution under challenging conditions.

4–6 h

14. Prepare to Walk Through a Complete GNSS-Dependent System Development from Requirements Through Verification

Skill: End-to-end system development ownership at the level a senior engineer is expected to demonstrate

Objective: Senior GNSS engineers at aviation integrators, defence prime contractors, and space agencies are expected to own technical threads from system requirements through algorithm design through verification testing through certification evidence. For an aviation GNSS application, this means understanding not just how RAIM works but how the algorithm's protection level computa-

tion is verified against the DO-229 compliance matrix, how the test cases are selected for DO-178C software qualification, and how the system performance is demonstrated against the TSO availability and integrity requirements across a representative set of satellite geometries and fault scenarios. The interview at this level is a peer discussion — but the peer discussion requires a genuine system development story, not a theoretical account.

4–6 h

Done Signal: For one GNSS-dependent system you personally owned the algorithm or architecture for — a receiver, an augmentation system, an integrated navigation system — you can discuss without notes: the originating operational requirement, the key algorithm or architecture trade-off that shaped the design, the verification approach, the hardest anomaly found during system testing, how it was resolved, and what changed in the algorithm or its parameters as a result.

15. Prepare a Technically Grounded Position on LEO Navigation Augmentation — What It Changes, What It Requires, and What Is Not Yet Solved

4–5 h

Skill: Next-generation GNSS architecture technical awareness at senior engineering depth

Objective: Low Earth Orbit navigation augmentation — Xona Space Systems (Pulsar), Satelles (STL on Iridium), and the OneWeb/Telesat proposals — represents the most significant architectural development in GNSS since the introduction of L5/E5. LEO satellites produce large Doppler rates but also rapid geometry change and high signal power (20–30 dB above GPS at the surface). This enables fast convergence for PPP and jamming resistance through higher received power. It also introduces new challenges: the fast Doppler rate stresses tracking loop design, the orbit determination accuracy of commercial LEO constellations is unproven at navigation levels, and the integrity characterisation of a LEO-augmented solution requires new fault models that do not yet exist in regulatory frameworks.

Done Signal: You have a prepared 3-minute technical discussion covering one specific engineering challenge in LEO navigation augmentation — named specifically: tracking loop design for high Doppler rate signals, LEO orbit determination accuracy requirements for navigation-grade positioning, integrity monitoring for a combined MEO/LEO solution, or antenna design for simultaneous MEO and LEO reception. Your position is technically grounded and connects your engineering background to the challenge rather than describing it from the outside.

16. Prepare to Lead a Technical Discussion on Filter Design Trade-Offs for a Specific Application Scenario

4–5 h

Skill: Architectural decision-making under competing requirements at senior engineering speed

Objective: Senior GNSS engineers are expected to make defensible architectural choices without time to run a simulation. “For an autonomous vehicle operating in an urban canyon environment, where GNSS availability varies between 2 and 8 satellites visible, and the IMU is a tactical-grade MEMS unit with 0.5°/hr gyro bias instability, should you implement loose or tight coupling? What process noise model would you use for the IMU error states? How would you handle a GNSS outage longer than 30 seconds?” These questions do not have a single correct answer — they have a correct analytical process and a justifiable conclusion.

Done Signal: You have worked through at least three filter design trade-off problems in your specific application domain — automotive, aviation, or maritime — stating your assumptions, reasoning through the trade-off, reaching a specific conclusion, and identifying what you would need to change if one assumption were different. You can do this conversationally, without a whiteboard, in under 5 minutes per problem.

17. Understand the PPP-RTK Convergence Mechanism and the State of the Art in Its Operational Deployment 4–5 h

Skill: Strategic precision positioning technology awareness at senior engineering depth

Objective: PPP-RTK — Precise Point Positioning with Integer Resolution, combining the global accessibility of PPP with the fast convergence of network RTK — is the precision positioning technology with the highest commercial momentum in 2025–2026. Hexagon’s Terra Office, Trimble’s RTX, and Galileo’s High Accuracy Service are all variants. The core technique transmits satellite clock and orbit corrections, satellite phase biases, and ionospheric grid corrections sufficient to resolve carrier phase ambiguity with reference receiver infrastructure — enabling centimetre-level positioning with 30–60 second convergence globally. Senior engineers who understand what the satellite phase bias corrections represent, how the network ionospheric corrections are generated, and what limits convergence time are specifically valuable to precision agriculture, autonomous vehicle navigation, and survey instrument developers.

Done Signal: You can explain what “phase biases” in PPP-RTK mean — why satellite and receiver hardware phase offsets prevent ambiguity resolution in standard PPP — and how the reference network’s ambiguity resolution enables the computation of integer-consistent corrections that allow the rover to resolve ambiguities. You can state what limits convergence time in the absence of ionospheric corrections and what specific benefit the ionospheric grid adds. You are aware of at least one operational PPP-RTK service and can characterise its performance specification.

18. Build Quantitative Estimation Fluency for GNSS System Performance — Without Running a Simulation 4–5 h

Skill: Senior-speed system-level reasoning about accuracy, availability, and integrity

Objective: Senior GNSS engineers are called on to sanity-check performance claims from suppliers, estimate whether a proposed architecture can meet a specification before committing to a design, and respond credibly to “does this make sense?” in programme reviews — often in real time. For a positioning application: estimating position accuracy from UERE and PDOP without running a simulation. For an integrity application: estimating RAIM availability for a given fault probability and satellite geometry. For an augmentation application: estimating PPP convergence time from tropospheric and ionospheric error levels. For an interference scenario: estimating the maximum jamming range for a given jammer power and receiver antenna gain. None of these requires a full simulation to estimate to order-of-magnitude if domain fundamentals are solid.

Done Signal: You have worked through five system performance estimation problems in your specific domain — stated out loud, assumptions named before numbers — arriving at order-of-magnitude answers with stated uncertainty. You have identified the most load-bearing assumption in each and described how you would verify it. Problems span at least two of: accuracy estimation from UERE and geometry, RAIM availability estimation, convergence time estimation for PPP, jamming range estimation, and IMU drift accumulation during a GNSS outage.

All Levels

19. Prepare a Specific Answer to “Why GNSS — and Why This Specific Application Area?” That Reflects Genuine Technical Engagement 1–2 h

Skill: Institution-specific motivation that distinguishes field-committed candidates from generic applicants

Objective: GNSS is a niche field. Employers — particularly specialist receiver manufacturers, SBAS developers, and precision instrument companies — are not looking for engineers who find navigation “interesting.” They are looking for engineers who have engaged with a specific aspect of GNSS

1–2 h

deeply enough to have a technical opinion about it: what makes PPP-RTK convergence faster than traditional PPP and slower than network RTK, what the transition to triple-frequency civil signals enables for single-frequency legacy receivers, why the Galileo E5 AltBOC signal has the best pseudo-range noise performance of any current civil signal. The expected answer in 2025–2026 engages with one of: the precision positioning technology transition underway (PPP-RTK, LEO augmentation), the aviation safety use case (ARAIM enabling global LPV-200 approaches), the autonomous vehicle GNSS robustness challenge (urban environment, spoofing, high-integrity low-latency requirement), or the defence assured PNT challenge (anti-jam, anti-spoof, M-code).

Done Signal: You have a 90-second answer that names your target application domain, identifies one specific current engineering challenge in that domain, explains what makes it technically difficult (not just commercially important), and connects your background to contributing to its solution. The answer cannot be given unchanged to a candidate for a general avionics or aerospace systems role.

1–2 h

20. Map Your Technical Background to a Specific GNSS Employer Type, Role Level, and Product Line — and Verify Security Clearance Eligibility

Skill: Precise self-positioning in a diverse employer landscape

Objective: The GNSS industry spans employers as different as a 10-person software receiver startup and a 100,000-person defence prime contractor. The algorithm engineer at Swift Navigation works in Python and C++ on an autonomous vehicle positioning problem. The GNSS engineer at Raytheon works in Ada or C++ on a classified military receiver with access requirements that begin before the offer is made. The applications engineer at Trimble supports field surveyors using RTK equipment in demanding environments. The research engineer at ESA's ESTEC Navigation Laboratory in Noordwijk evaluates Galileo signal-in-space performance. These roles share a technical vocabulary but differ in day-to-day work, interview format, clearance requirements, programming language expectations, and career trajectory.

Done Signal: You can state in under 60 seconds: the GNSS employer type you are targeting (receiver manufacturer, defence prime, aviation integrator, chipset vendor, space agency, precision agriculture platform, or survey instrument company), the specific technical domain within GNSS that is your primary strength, the programming languages and tools you work in fluently, your security clearance status or eligibility in your target country, and two specific technical contributions your background makes to the target role.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	Implement pseudorange PVT solver from scratch — Python, real data	All	8–12 h
2	Four-constellation knowledge — signals, frequencies, orbit types	All	4–5 h
3	Kalman filter — derive update equations, not just apply them	All	4–6 h
4	STAR stories — diagnosis, cross-discipline, quality, failure	All	2–3 h
5	GNSS error budget — all sources, quantitative magnitudes	All	3–4 h
6	Hands-on real GNSS data — RINEX, CORS, u-blox, or Android	All	4–6 h
7	SBAS architecture — monitor station through airborne receiver	Mid+	6–8 h
8	INS/GNSS integration — loose, tight, and deep coupling	Mid+	5–6 h
9	Carrier phase and integer ambiguity resolution — RTK, PPP-AR	Mid+	5–7 h
10	GNSS system failure diagnosis story — data-driven root cause	Mid+	2 h
11	Interference — jamming, spoofing, and engineering mitigations	Mid+	4–5 h
12	RAIM and ARAIM — protection level computation, integrity trade-offs	Mid+	5–6 h
13	Live coding readiness — RINEX parsing, satellite positions, WLS	Mid+	6–8 h
14	Complete system development story — requirements through verification	Senior	4–6 h
15	LEO navigation augmentation — technical position on one open problem	Senior	4–5 h
16	Filter design trade-off discussion — conversational, in real time	Senior	4–5 h
17	PPP-RTK — convergence mechanism and operational deployment status	Senior	4–5 h
18	System performance estimation — five problems, stated out loud	Senior	4–5 h
19	“Why GNSS — why this application area?” — technically specific	All	1–2 h
20	Employer type / domain / clearance self-positioning	All	1–2 h

THE STRUCTURAL REALITY

GNSS is a field where the hardest problems are simultaneously invisible and ubiquitous. The signals are received at power levels twenty decibels below thermal noise floor — you cannot directly observe them without despreading. The position errors that matter most are the rare, large ones — the ones that kill aircraft or misdirect autonomous vehicles — not the everyday centimetre-level noise. The ionosphere is a dynamic, partially correlated, partially observable disturbance source that has been studied for fifty years and is still not fully modelled. Integer ambiguity resolution is an NP-hard combinatorial problem that practical receivers solve in real time using clever heuristics and careful covariance modelling.

The engineers who succeed in GNSS are those who are genuinely interested in the inverse problem at its core: inferring precise position, time, and attitude from the statistical aggregation of noisy, correlated, occasionally faulty radio measurements from satellites whose positions are imperfectly known. The field rewards mathematical rigour, careful thinking about error statistics, and the kind of patient data analysis that turns an anomalous residual into a root-cause understanding.

The engineers who struggle are those who treat GNSS as a commodity — a positioning service you call when you need coordinates. GNSS is not a service. It is an inference problem, and every design decision in a GNSS system is ultimately a decision about how to solve that inference problem better, faster, more reliably, or under more adverse conditions than the system that existed before.

BREAK INTO AEROSPACE
